Practice Set 19 SAQ Sheet – Short Answer Practice

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Short-answer practice sheet compiled from handwritten study lists.

Q3a. Sterilisation of urological instruments

Type: SAQ (10 marks)

Definitions

- **Cleaning:** removes debris that shields organisms; always first.
- **Disinfection:** kills most microbes but not all spores. HLD (high-level disinfection) kills all except large numbers of spores.
- **Sterilisation:** kills all life, including spores.

Spaulding classification

Category	Contact	Requirement	Urological examples
Critical	Sterile tissue, vessels	Sterilisation	Resectoscope, ureteroscope, nephroscope, laparoscopic and robotic instruments
Semi-critical	Mucosa	HLD at least (sterilise if possible)	Flexible cystoscope, TRUS (transrectal ultrasound) probe
Non-critical	Intact skin	Low-level disinfection	Abdominal ultrasound probe

Reprocessing steps

1. Bedside pre-clean; leak test for flexible scopes.
2. Manual cleaning with enzymatic detergent; brush and flush channels.
3. Rinse → disinfect or sterilise → rinse → dry channels (alcohol, air) → store hanging; record tracking.

Methods

Method	Conditions	Use and drawbacks
Steam autoclave	121 °C for 15 min (15 psi) or 134 °C for 3–4 min	Cheapest, most reliable; metal instruments, autoclavable optics
Dry heat	160 °C for 2 hours	Glass; rarely used
Ethylene oxide	Low temperature	Heat-sensitive scopes, cables; toxic, carcinogenic, long aeration
Hydrogen peroxide gas plasma	Low temperature, short cycle	No toxic residue; not for cellulose; long narrow lumens limited
Peracetic acid (Steris)	Automated liquid, low temperature	Sporicidal; ureteroscopes
Glutaraldehyde 2%	HLD in 20 min; sterilant after about 10 hours	Cheap; irritant, sensitiser; must rinse well
Gamma irradiation	Industrial	Single-use catheters, stents, sutures

Monitoring

- **Physical:** time, temperature, pressure records.
- **Chemical indicators:** autoclave tape; Bowie–Dick test (air removal).
- **Biological indicators:** *Geobacillus stearothermophilus* spores (steam, plasma); *Bacillus atrophaeus* (ethylene oxide, dry heat).

Urology-specific points

- A laser-perforated ureteroscope channel lets fluid in during sterilisation, destroying the scope.
- Poor channel cleaning causes outbreaks (Pseudomonas, mycobacteria).
- **Prions** resist routine methods: single-use or special protocols.
- **Single-use scopes** avoid reprocessing (about $107 per case for reusable ureteroscopes).
- **Also know:** laser fibre re-sterilisation; limited-life robotic instruments.

Diagram to draw: reprocessing cycle flowchart.

> **Key points to remember:**
>
> - Spaulding: critical sterilise, semi-critical HLD, non-critical low-level.
> - Autoclave 121 °C 15 min or 134 °C 3–4 min.
> - Ureteroscopes: ethylene oxide, glutaraldehyde or peracetic acid.
> - Biological indicator: *G. stearothermophilus*.

Q3b. Post-obstructive diuresis

Type: SAQ (10 marks)

Definition

- POD (post-obstructive diuresis): marked polyuria after relief of obstruction, as the kidney sheds retained salt, water and urea.
- No agreed rate; a common working definition is **>200 mL/h for 2 consecutive hours or >3 L in 24 hours**. Up to 300 mL/h is common.
- **Physiological** (appropriate, self-limited) versus **pathological** (continues after euvolaemia, causing hypovolaemia and electrolyte loss).

Incidence and setting

- Unilateral obstruction: **about 2%** (the other kidney compensates). Bilateral or solitary kidney: **about 50%**.
- Commonest after catheterising **chronic retention with renal impairment**; also bilateral ureteric obstruction and PUV (posterior urethral valves).

Mechanisms

1. **Transporter dysregulation:** down-regulation of Na^+/K^+-ATPase and aquaporin-2 → concentrating defect (nephrogenic DI (diabetes insipidus)); loss of medullary tonicity.
2. **Volume expansion:** on relief, raised ANP (atrial natriuretic peptide), low aldosterone and sympathetic tone → salt and water loss.
3. **Osmotic diuresis** from retained urea.

Risk factors

- Bilateral obstruction, raised creatinine, oedema, heart failure, hypertension, confusion.
- After valve ablation: oligohydramnios, preterm birth, raised creatinine.

Clinical features

- Starts within hours of relief; most settle within 24–48 hours.
- Danger signs: postural hypotension, tachycardia, weight loss, hypernatraemia (free-water loss), low potassium, magnesium, phosphate.

Management

```
Relieve obstruction (catheter / nephrostomy)
        |
Monitor hourly urine output, vitals, weight
        |
Alert, can drink -> free oral fluids (most resolve)
        |
Output persists / cannot drink / unstable
  -> admit; electrolytes every 4-6 h
  -> intravenous 0.45% saline: about 75% of urine output
  -> replace K, Mg, phosphate; adjust to urine sodium/osmolality
```

- **Do not overcorrect:** replacing every millilitre sustains the diuresis.
- **Gradual bladder decompression has no role**: it does not prevent haematuria or diuresis.
- PCN (percutaneous nephrostomy) for upper-tract obstruction; hydration alone for physiological POD.

- **Also know:** decompression haematuria; neonatal PUV concentrating defect; no approved drug therapy.

Diagram to draw: graph of urine output against time after catheterisation.

> **Key points to remember:**
> - Unilateral about 2%, bilateral about 50%.
> - Mechanisms: aquaporin-2 loss, volume expansion, urea osmotic diuresis.
> - Replace about 75% of output with 0.45% saline; do not overcorrect.
> - Gradual decompression has no role.

Q3c. Renal mucormycosis

Type: SAQ (10 marks)

Introduction

- Mucormycosis: an **angioinvasive** fungal infection by Mucorales (*Rhizopus* commonest; *Mucor*, *Lichtheimia*).
- Kidney involvement occurs in **disseminated** disease, or as **isolated renal mucormycosis**, reported particularly from India, sometimes in apparently immunocompetent young adults. Cases rose during the 2021 COVID-19 (coronavirus disease 2019) wave.

Risk factors

- Uncontrolled diabetes, especially **ketoacidosis**; corticosteroids; neutropenia and haematological cancer; transplantation; iron overload or deferoxamine; intravenous drug use; AIDS (acquired immunodeficiency syndrome).

Pathogenesis

- Spores inhaled or ingested → **haematogenous** seeding of the kidney.
- Hyphae invade arteries and veins → **thrombosis → infarction and haemorrhagic necrosis**.
- Acidosis and free iron favour growth (hence the ketoacidosis risk).

Clinical features

- Fever, flank pain, tender flank mass, haematuria, pyuria with sterile bacterial cultures.
- **Bilateral** disease → AKI (acute kidney injury) or anuria.
- Suspect when "pyelonephritis" fails antibiotics in a high-risk patient.

Investigations

- **CECT (contrast-enhanced computed tomography):** enlarged kidney(s), wedge-shaped or diffuse **non-enhancing** areas (infarcts), perinephric stranding or collection, renal vessel thrombosis.
- **Tissue diagnosis** (biopsy or nephrectomy specimen): **broad, ribbon-like, aseptate or pauciseptate hyphae branching at right angles**, invading vessels.
- KOH (potassium hydroxide) mount, calcofluor stain, fungal culture (often negative), PCR (polymerase chain reaction).
- Differential: *Aspergillus* has narrow septate hyphae with acute-angle branching.
- Image sinuses, lungs, brain.

Management (three pillars)

1. **Reverse predisposition:** control glucose, correct ketoacidosis, taper steroids and immunosuppression, stop deferoxamine.
2. **Early surgical debridement:** nephrectomy for a unilateral necrotic kidney; drain collections.
3. **Antifungal: liposomal amphotericin B 5 mg/kg/day** first line (ECMM/MSG ERC (European Confederation of Medical Mycology/Mycoses Study Group Education and Research Consortium) guideline, 2019). Step-down or salvage: **posaconazole or isavuconazole**.

- **Fluconazole, voriconazole and echinocandins are inactive.**
- Bilateral: antifungal plus dialysis.

Prognosis

- Mortality is high, especially with bilateral or disseminated disease and delayed diagnosis.

Diagram to draw: right-angle branching aseptate hypha invading a renal vessel.

> **Key points to remember:**
>
> - Angioinvasion → infarction; diabetic ketoacidosis is the key risk.
> - CECT: non-enhancing enlarged kidney; histology: broad aseptate right-angle hyphae.
> - Liposomal amphotericin B 5 mg/kg/day plus nephrectomy and glucose control.
> - Voriconazole and echinocandins do not work.

Q3d. Hydatid disease of kidney

Type: SAQ (10 marks)

Introduction

- Cystic echinococcosis (CE) by *Echinococcus granulosus*, a cestode. Dog: definitive host; sheep and humans: intermediate.
- Ingested eggs → liver (commonest), lung, then **kidney (third site, about 2–3%)** via the systemic circulation. Usually single, unilateral, cortical.

Pathology

- Wall: **pericyst** (host fibrous layer), **laminated membrane** (ectocyst), **germinal layer** (endocyst) producing brood capsules, scolices and daughter cysts ("hydatid sand").
- **Closed** versus **open** cyst (ruptured into the collecting system → **hydatiduria**).

Clinical features

- Grows 1–2 cm a year; long asymptomatic; flank mass, dull pain.
- **Hydatiduria** ("grape-skin" membranes in urine; pathognomonic) with colic, in a minority.
- Infection; rupture (anaphylaxis). Renal function usually preserved.

Investigations

- **Serology** (ELISA (enzyme-linked immunosorbent assay)) is adjunctive only (sensitivity 60–90%); eosinophilia; Casoni test obsolete.
- **Ultrasound and CT (computed tomography):** thick-walled cyst, wall calcification, daughter cysts, detached membranes ("water-lily sign"). Stage guides treatment.

WHO (World Health Organization) stage	Appearance	Activity
CE1	Unilocular, double-line wall	Active
CE2	Multivesicular, daughter cysts	Active
CE3a / CE3b	Detached membranes / daughter cysts in solid matrix	Transitional
CE4 / CE5	Solid "ball of wool" / calcified wall	Inactive

- Retrograde pyelogram shows communication; image liver and lungs. **Never aspirate blindly** (anaphylaxis, seeding).

Management

- **Albendazole 400 mg twice daily**, 1–6 months: alone for small or inoperable cysts; around surgery to reduce seeding (± praziquantel).
- **Surgery is the mainstay** (open, laparoscopic or robotic), nephron-sparing when possible:
  1. Pack the field with scolicide-soaked swabs (hypertonic saline, povidone-iodine).
  2. Inject scolicide only if **not** communicating (chemical injury to the collecting system).
  3. **Pericystectomy, partial nephrectomy or deroofing**; close communications; omentoplasty or drain.
  4. **Nephrectomy** for a non-functioning kidney.
- **PAIR (puncture, aspiration, injection, re-aspiration):** chapter says liver only, not kidney; some series report percutaneous renal treatment.
- Calcified inactive cysts (CE4, CE5) may be observed.

Diagram to draw: layers of a hydatid cyst with daughter cysts and a communication into a calyx.

Key points to remember:

- Kidney: third site (about 2–3%).
- Hydatiduria (grape-skin membranes) is pathognomonic of a communicating cyst.
- Albendazole 400 mg twice daily around surgery; no scolicide if communicating.

Q3e. Ethics and informed consent in urology

Type: SAQ (10 marks)

Four principles (Beauchamp and Childress, 1979)

Principle	Meaning
Autonomy	Patient's deliberated self-rule; basis of consent and confidentiality
Beneficence	Act for the patient's benefit, judged by the patient's values
Non-maleficence	Avoid harm (*primum non nocere*); requires training and honest risk data
Justice	Fair distribution of resources, benefits and risks

- Hippocratic Oath gave beneficence, non-maleficence, confidentiality; autonomy and justice are modern additions (autonomy is bounded by justice).
- **Jonsen's four-box method:** medical indications; patient preferences; quality of life; contextual features (family, cost, law).

Informed consent: legal evolution

- From battery to negligence. **Schloendorff (1914):** right to bodily integrity. **Salgo (1957):** term "informed consent". **Canterbury v Spence (1972): reasonable-patient** disclosure standard.
- India: **Samira Kohli (2008)**: consent for one procedure does not cover another, except in emergency.

Elements

1. Diagnosis, proposed treatment and alternatives, including no treatment.
2. Risks, benefits and uncertainty compared with alternatives.
3. **Check understanding** (most often missed).
4. Voluntary decision by a patient with capacity; documented.

- Material risks: no agreed cut-off (1% proposed); disclose rarer ones by severity; lay language.
- Urology examples: incontinence and erectile dysfunction after prostatectomy; retrograde ejaculation after TURP (transurethral resection of the prostate); stent symptoms; conversion to open surgery.

Exceptions

- **Emergency** in an incapable patient (presumed consent); a conscious patient may still refuse.
- **No capacity:** surrogate using substituted judgement, then best interests.
- **Minors (<18 years):** parents consent; child gives assent.

Improving consent

- Decision aids, repeat-back, simple written material.
- **Shared decision making:** two-way exchange of options and patient values; vital when several options are reasonable (e.g. localised prostate cancer).

Special areas

- **End of life:** early goals-of-care talks, advance directives, DNR (do-not-resuscitate) orders.
- **Research:** written consent, ethics committee approval, extra protection for vulnerable groups.
- **Also know:** confidentiality; conflicts of interest; futility policies.

Diagram to draw: Jonsen four-box grid.

Key points to remember:

- Four principles: autonomy, beneficence, non-maleficence, justice.
- Canterbury v Spence: reasonable-patient standard; Samira Kohli in India.
- Commonest gap: not checking understanding.
- Exceptions: emergency, incapacity (surrogate), minors.

Q3f. Operating room set-up

Type: SAQ (10 marks) (see also 2021-Summer Paper 4, Q3a for operating room safety)

Physical design

- **Zones:** protective (reception, changing) → clean (pre-op, recovery) → sterile (theatre, scrub) → disposal; one-way flow.
- **Ventilation:** positive pressure, about 20 air changes an hour, filtered air; laminar flow for implants.
- **Environment:** temperature about 20–24 °C, relative humidity about 20–60%. Patient warming: even 1–2 °C hypothermia raises SSI (surgical site infection) and blood loss (**Kurz 1996**: 3-fold wound infection).
- Power backup, piped gases, anaesthetic scavenging, radiolucent table.

Equipment layout by procedure

Setting	Arrangement
Endourology (TURP (transurethral resection of the prostate), URS (ureteroscopy), PCNL (percutaneous nephrolithotomy))	Lithotomy or prone; tower and monitor directly ahead; C-arm opposite surgeon; wires, baskets, dilators, spare scopes ready
Laparoscopic, upper abdomen	Surgeon and assistant face the abdomen; monitor opposite; scrub nurse near the foot
Laparoscopic, pelvic	30° Trendelenburg, split legs; surgeon on preferred side, assistant opposite
Robotic	Patient cart, surgeon console, vision cart; Si docks between legs, Xi from the side

- **Checklist before laparoscopic incision:** irrigation–suction, electrosurgical unit, full CO_2 cylinder plus spare, white-balanced camera, insufflator and Veress needle checked, **open tray ready**.
- Cables fixed; camera cable apart from diathermy; insufflator at eye level.

Positioning safety

- Pad bony prominences; axillary roll (lateral); avoid prolonged kidney rest and full flexion (rhabdomyolysis); no shoulder braces in steep Trendelenburg (brachial plexus).

Ergonomics and staff protection

- Monitor straight ahead, top at eye level; elbows 90–120°; table at the surgeon's pubic symphysis (laparoscopy).
- **Radiation:** ALARA (as low as reasonably achievable): time, distance, shielding; lead apron, thyroid shield, glasses; dosimeter.
- Smoke evacuation; sharps safety; vaccination.

Safety systems

- **WHO (World Health Organization) Surgical Safety Checklist** (sign in, time out, sign out); counts; fire-risk assessment (oxidiser, fuel, ignition).

Diagram to draw: theatre plan: table, tower, C-arm, anaesthetist, surgeon, scrub nurse.

Key points to remember:

- Zones with one-way flow; positive pressure, about 20 air changes an hour.
- Monitor directly ahead at eye level; open tray ready for laparoscopy.
- Hypothermia triples wound infection (Kurz).
- ALARA: time, distance, shielding.

Rapid Revision Sheet – Practice Set 19 SAQ Sheet

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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